Empirical IO – Problem Set 2: Demand and Supply Estimation

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1 Part I: Logit Demand

Cleaning. Note that 63 Canadian apps appeared twice in the dataset, each time with a different estimate of downloads. As we had no way to know which estimate is correct, we dropped all Canadian observations and excluded Canada from the estimation. Our results are very similar if we instead keep all Canadian observations, only the lower-ranked duplicates, or only the higher-ranked ones.

1.1 Berry Logit

We use two instruments and their combination: first, as suggested by the problem set, we use variation in the number of competing apps in the same genre and country (count). Second, we use the share of other apps with in-app purchases in the same genre and country (other IAP). Then, we combine these instruments. Additionally, in all specifications we control for the average rating score and in-app purchases. For robustness, we introduce country fixed effects as controls to the specification with combined instruments. The interpretation is that we control for the fact that different markets have different propensities to choose the outside option. Note that we excluded outside apps from all estimations.

We estimate

$$\ln(s_{jc}) - \ln(s_{0c}) = \gamma\, p_{jc} + \beta_1\, \text{IAP}_{jc} + \beta_2\, \text{Score}_{jc} + \xi_{jc}, \tag{1}$$

where s_{jc} is the download share of app j in country c, s_{0c} is the share of the outside option, and ξ_{jc} is unobserved quality. Since p_{jc} is likely correlated with ξ_{jc}, we instrument it.

In Table 1, we report the results from the Berry logit estimates.

Table 1: Berry logit: demand estimates

	(1) OLS	(2) IV: count	(3) IV: other IAP	(4) IV: both	(5) IV: both + country FE
Price	0.0040	-2.0887	-1.5567**	-1.7000***	-1.4500***
	(0.0108)	(1.3213)	(0.6653)	(0.6516)	(0.5157)
IAP control	✓	✓	✓	✓	✓
Average score control	✓	✓	✓	✓	✓
Country fixed effects					✓
Observations	4158	4158	4145	4145	4145
R^2	0.070				

Robust standard errors in parentheses.

$*p < 0.10,\ **p < 0.05,\ ***p < 0.01$.

When using OLS, we obtain a price coefficient that is essentially zero (and slightly positive). This is to be expected, given that price is an endogenous variable, so the OLS estimates are not valid. For example, unobserved quality is not controlled for, and it is likely to be positively correlated with both prices and quantities, so the price coefficient is biased upwards. Note that the coefficient being positive is unreasonable as it would mean there is no tradeoff to increasing prices, since it would increase both revenue per consumer and the number of consumers. For the estimates to be valid, we need instruments: the estimates we obtain this way are consistent and lie in a relatively narrow range. Given the similarity of our estimates, we can plausibly conclude that the IVs employed are valid. The estimates stay similar when adding country fixed effects.

Note that the number of observations is lower when we use the other-IAP instrument, as it excludes apps that had no other apps in the same genre and country. However, this excludes only 13 apps, so the effect is likely to be marginal.

Table 2: First-stage regressions: dependent variable is price

	(1) Count	(2) Other IAP	(3) Both	(4) Both + country FE
Competing apps	0.0062*		0.0044	0.0081**
	(0.0037)		(0.0037)	(0.0039)
Other apps' IAP		-0.5755***	-0.5234**	-0.4121*
		(0.2213)	(0.2240)	(0.2281)
IAP	-0.1759*	-0.1108	-0.1071	-0.1002
	(0.1060)	(0.1163)	(0.1160)	(0.1155)
Average score	-0.1473	-0.1168	-0.1186	-0.1340
	(0.0908)	(0.0910)	(0.0909)	(0.0911)
Constant	4.4761***	4.6362***	4.5211***	4.3992***
	(0.4109)	(0.3896)	(0.4114)	(0.4317)
Country fixed effects				✓
Observations	4158	4145	4145	4145
R^2	0.002	0.003	0.003	0.012
F-value	2.88	6.76	3.99	4.87

Robust standard errors in parentheses.

$*p < 0.10,\ **p < 0.05,\ ***p < 0.01.$

In the first stage, the number of competing apps is associated with higher prices (here, the coefficient should be interpreted as the impact of one more competitor), while the share of other apps offering in-app purchases is associated with lower prices (here, the coefficient should be interpreted as going from no competitors to all competitors in the same category offering in-app purchases). See the results reported in Table 2.

Note that when using both instruments, the first-stage estimates do not change substantially. Also note that these instruments are not particularly strong, as the F-test values are not very high.

1.2 Nested Logit

For the nested logit, we instrumented both price and the within-group market share, hence the need for two instruments.

Table 3: Nested logit: demand estimates

	(1) OLS	(2) IV: both	(3) IV: both + country FE
Price	0.0051	-1.7495***	-1.5133***
	(0.0052)	(0.6700)	(0.5547)
Within-nest log share	0.8851***	0.1068	-0.1630
	(0.0073)	(0.2337)	(0.2248)
IAP control	✓	✓	✓
Average score control	✓	✓	✓
Country fixed effects			✓
Observations	4158	4145	4145
R^2	0.786		

Robust standard errors in parentheses.

IV instruments both price and within-nest log share using competing-app count and other apps' IAP.

$*p < 0.10,\ **p < 0.05,\ ***p < 0.01.$

In the OLS model, the price coefficient is not significant and very close to zero, as in the Berry logit model, while the nesting coefficient is positive and highly significant. This is to be expected, given that apps with a larger within-nest share mechanically have a larger market share as well. The IV specification, used to correct for the endogeneity of both price and the within-nest log share, yields results similar to the IV estimation of the first model, with a non-significant nesting coefficient. Moreover, the nested coefficient changes sign when we control for country fixed effects, meaning that the within-nest estimate is particularly unstable (and a negative sign is not reasonable). Therefore, it is probably safe to proceed with the simple logit model, as we didn't find evidence for strong within-nest correlation.

2 Part II: Supply Analysis

2.1 Cost Estimation

First, note that we used our estimates from the 11-country sample (as we excluded Canada) to estimate costs on the US sample. Second, one developer owned both an outside app and an inside app. Since we normally think of the outside good as not downloading any of the apps under consideration, we treated the outside apps as not belonging to any developer (they were also excluded from the estimation).

All the derivations of the cost and markup formulas are in the Appendix.

We report our results for the current ownership structure in Tables 4 and 5; see the discussion in the next section.

2.2 Markup Estimation

Table 4: Average markups under alternative ownership structures

	OLS	IV: count	IV: other IAP	IV: both	IV: both + Country FE
Current ownership	-248.400	0.481	0.646	0.591	0.694
Single-product firms	-247.571	0.480	0.644	0.590	0.691
Monopolist	-2293.534	4.445	5.964	5.461	6.403

Mean absolute markup (price minus marginal cost).

Unweighted averages across inside apps in the US.

Observed prices and market shares are held fixed.

For each ownership structure we have a different markup formula. For the current ownership structure, it is $\mu_j = -1/[\gamma(1 - S_f)]$, with S_f being the sum of the market shares of all products owned by the owner of product j. For single-product firms, we use s_j, the market share of product j, instead of S_f. In the monopolist scenario, the monopolist owns all inside apps, so $1 - S_f = s_0$ and $\mu_j = -1/(\gamma s_0)$: that is, we assume that outside apps are not owned by the same developer. Otherwise, we would reach the impossible situation in which the developer could raise prices as high as it wishes, as we normally think of the outside option as not downloading any of the apps.

We obtain extremely large negative markups when considering the OLS specification. The reason is that the OLS price coefficient γ is very small but positive (Table 1), yielding large and negative markups. Of course, these estimates are unreasonable, no producer would have large negative markups, but it is also unreasonable to have a positive coefficient on prices. This could be due to endogenous prices: high unobserved quality probably raises both demand and prices, making demand appear less price-sensitive than it actually is. Since $c_j = p_j - \mu_j$, the negative OLS markups in turn imply implausibly large marginal costs (Table 5). By contrast, the markups estimated through the IV specifications are similar to each other within each ownership structure.

When looking at the specifications that include instrumental variables, we obtain more reasonable magnitudes both for the current ownership structure and for single-product firms, with the former always having slightly higher margins: as owners control a higher market share and more products, they can internalize the effect of a price increase of product j on their other products. The third ownership structure, i.e. the "Monopolist" row of Table 4, sees a stark increase in markups, as the monopolist can internalize the effect of a price increase of product j on all the other products, leading to much higher markups across all apps. Under all IV specifications, these markups imply negative average marginal costs (Table 5). Notably, in performing this exercise, we assumed the

Table 5: Average marginal costs under alternative ownership structures

	OLS	IV: count	IV: other IAP	IV: both	IV: both + Country FE
Current ownership	252.388	3.507	3.342	3.397	3.295
Single-product firms	251.559	3.508	3.344	3.399	3.297
Monopolist	2297.522	-0.457	-1.976	-1.473	-2.415

Mean marginal cost

Unweighted averages across inside apps in the US.

Observed prices and market shares are held fixed.

apps to be owned by a monopolist, when in reality they are not. Given the estimates of negative marginal costs, these prices are not realistic: nevertheless, we assume these would have been the optimal prices of monopolists under the monopolist scenario. There are no large differences between current ownership and single-product firms, as the current ownership structure is close to the single-product one. See Tables 4 and 5 for the estimated average markups and marginal costs.

Adding country fixed effects result in slightly larger markups and slightly smaller costs, but they are in line with the other IV estimates.

2.3 Merger Simulation

In this last part, we considered a scenario in which the market leader Mojang considers acquiring another developer. We simulated the Bertrand–Nash equilibrium in which firms are asymmetric competitors selling differentiated products. For the merger simulation, we use the IV estimate with both instruments (competing-app count and other apps' IAP), with no country fixed effects.

We simulated the acquisition of each of the three largest developers other than Mojang, ranked by their combined US market shares. Under differentiated-product Bertrand competition, a merger allows the acquiring firm to internalize substitution between its own products and the acquired apps, potentially increasing combined profits even without cost efficiencies. We compared the three scenarios to identify the most profitable acquisition among the candidates considered.

Table 6: Equilibrium price changes for the merging firms

	Minecraft			Acquired apps
Target	Before	After	Change	Change per app
Rockstar Games	6.9900	7.0197	+0.0297	+0.1057
Fireproof Games	6.9900	7.0091	+0.0191	+0.1054
SQUARE ENIX Ltd	6.9900	7.0087	+0.0187	+0.1054

Notes: Prices are expressed in the dataset's price units. Marginal costs are held fixed.

First, we analyze the change in the price of the app sold by Mojang, *Minecraft*, after each of the three simulated mergers. In all three scenarios, its price increases, which can be explained by the internalization of diverted profits coming from the substitution of Minecraft with the app(s) sold by the respective merging party. The merger with Rockstar Games generates a greater increase in the price of Minecraft than the other two. With the simple logit model, consumers switch proportionally to market shares, and among the remaining firms, Minecraft has about the same share in all three scenarios (only the denominator is different). However, the price of Minecraft changes differently because the market shares of the acquired firms differ: the more switchers would move to the acquired firm, the safer it is to raise prices. Moreover, note that the price of the acquired apps changes substantially more: intuitively, many of their switchers would shift to Minecraft, which has a substantial market share (about 15.2%), so the merged firm would benefit from them, while few switchers from Minecraft would move to the acquired firm, given that the market shares of the acquired firms are much smaller (about 2.2–3.5%).

Table 7 shows that all three mergers reduce market share and revenue relative to the combined pre-merger outcomes of the two firms. This result highlights that the price increase drives some consumers to rival apps (loss in market share), and that the higher margin does not compensate for the loss in the number of sales (loss in revenue). Still, the reduction in variable costs (fewer units produced and sold) and the higher margin per unit

Table 7: Changes relative to Mojang and the target combined before acquisition

Target	Δ Market share (pp)	Δ Revenue	Δ Operating profit
Rockstar Games	-1.2346	-68.3728	+0.5538
Fireproof Games	-0.8187	-26.7195	+0.3368
SQUARE ENIX Ltd	-0.8014	-22.1998	+0.3284

Notes: Changes equal post-merger outcomes minus the two firms' combined pre-merger outcomes. Marginal costs remain unchanged; operating profit excludes fixed costs and acquisition payments.

more than compensate for these losses, making all three mergers profitable. Among the candidates considered, Rockstar Games generates the largest gains, so we would recommend acquiring Rockstar Games.

Appendix

In the Appendix we derive the formulas for the markup estimation.[1]

A Single-product firms: demand derivatives

Let s_j denote app j's market share and s_0 the outside-option share. Write the logit demand equation as

$$\ln(s_j) - \ln(s_0) = \gamma p_j + \beta_1 \,\mathrm{IAP}_j + \beta_2 \,\mathrm{Score}_j + \xi_j, \qquad \gamma < 0. \tag{2}$$

Holding other prices, characteristics, and unobserved demand terms fixed,

$$\frac{1}{s_j}\frac{\partial s_j}{\partial p_j} - \frac{1}{s_0}\frac{\partial s_0}{\partial p_j} = \gamma, \tag{3}$$

$$\frac{\partial s_j}{\partial p_j} = \frac{s_j}{s_0}\frac{\partial s_0}{\partial p_j} + \gamma s_j. \tag{4}$$

For another app $k \neq j$, the same differentiation gives

$$\frac{\partial s_k}{\partial p_j} = \frac{s_k}{s_0}\frac{\partial s_0}{\partial p_j}, \qquad k \neq j. \tag{5}$$

Since all shares sum to one, their derivatives sum to zero. Hence

$$\begin{aligned}
0 &= \frac{\partial s_0}{\partial p_j} + \sum_{k=1}^{J} \frac{\partial s_k}{\partial p_j} \\
&= \frac{\partial s_0}{\partial p_j} + \frac{\partial s_j}{\partial p_j} + \sum_{k \neq j} \frac{\partial s_k}{\partial p_j} \\
&= \gamma s_j + \frac{\partial s_0}{\partial p_j}\left(1 + \frac{s_j}{s_0} + \frac{\sum_{k \neq j} s_k}{s_0}\right) \\
&= \gamma s_j + \frac{\partial s_0}{\partial p_j}\left(1 + \frac{\sum_{k=1}^{J} s_k}{s_0}\right) \\
&= \gamma s_j + \frac{\partial s_0}{\partial p_j}\left(1 + \frac{1 - s_0}{s_0}\right) = \gamma s_j + \frac{1}{s_0}\frac{\partial s_0}{\partial p_j}.
\end{aligned}$$

Therefore, the outside-option and cross-price derivatives are

$$\boxed{\frac{\partial s_0}{\partial p_j} = -\gamma s_0 s_j}, \qquad \boxed{\frac{\partial s_k}{\partial p_j} = -\gamma s_k s_j \quad (k \neq j)}. \tag{6}$$

Substituting the outside-option derivative back into the own-share equation,

$$\frac{\partial s_j}{\partial p_j} = \frac{s_j}{s_0}(-\gamma s_0 s_j) + \gamma s_j = \boxed{\gamma s_j (1 - s_j)}. \tag{7}$$

[1] *Note on AI usage:* this was handwritten, and we used an LLM to transcribe the handwritten notes to LaTeX.

B Quantities and the single-product markup

With fixed market size M, quantity is $q_j = Ms_j$. Thus

$$\frac{\partial q_j}{\partial p_j} = M\frac{\partial s_j}{\partial p_j} = \gamma q_j(1 - s_j). \tag{8}$$

For a single-product firm, the pricing condition implies

$$\boxed{p_j - c_j = -\frac{q_j}{\partial q_j/\partial p_j} = -\frac{1}{\gamma(1 - s_j)}}. \tag{9}$$

C Common ownership and monopoly

Assume the firm owns the set of apps $\mathcal{J}_f$. Its total profit is

$$\pi_f = \sum_{k\in\mathcal{J}_f} (p_k - c_k)q_k. \tag{10}$$

For each app $j \in \mathcal{J}_f$, the first-order condition is

$$\frac{\partial \pi_f}{\partial p_j} = q_j + (p_j - c_j)\frac{\partial q_j}{\partial p_j} + \sum_{\substack{k\in\mathcal{J}_f \\ k\neq j}} (p_k - c_k)\frac{\partial q_k}{\partial p_j} = 0. \tag{11}$$

Let $\mu_k = p_k - c_k$. Substituting the logit derivatives and dividing by q_j,

$$1 + \gamma\mu_j(1 - s_j) - \gamma \sum_{\substack{k\in\mathcal{J}_f \\ k\neq j}} \mu_k s_k = 0. \tag{12}$$

Expanding $\mu_j(1 - s_j)$ and including app j in the sum gives

$$\mu_j = \sum_{k\in\mathcal{J}_f} \mu_k s_k - \frac{1}{\gamma}. \tag{13}$$

The right-hand side is identical for every app owned by the firm, so $\mu_j = \mu_k = \mu$. Consequently,

$$0 = 1 + \gamma\mu\left(1 - s_j - \sum_{\substack{k\in\mathcal{J}_f \\ k\neq j}} s_k\right). \tag{14}$$

$$\boxed{\mu = -\frac{1}{\gamma\left(1 - s_j - \sum_{\substack{k\in\mathcal{J}_f \\ k\neq j}} s_k\right)} = -\frac{1}{\gamma(1 - S_f)}}, \qquad S_f = \sum_{k\in\mathcal{J}_f} s_k. \tag{15}$$

Monopoly specialization. If all inside apps are owned by the same firm, $S_f = 1 - s_0$, so the preceding expression gives

$$\boxed{\mu = -\frac{1}{\gamma s_0}}.$$